

1 Simultaneous gap inference, honest tuning, and contamination

1.1 The identifiable gap and an estimable baseline

Let $U \in [0, 1]^p$ have uniform one-dimensional marginals, with arbitrary cross-coordinate dependence. The structural tail-index surface is only identified on $\text{supp}(U)$, so the global baseline is

$$\gamma_U^* = \text{ess sup } \gamma(U) = \max_{x \in \text{supp}(U)} \gamma(x). \quad (1)$$

For a pointwise conditional support $C_j(u) = \text{supp}\{U_{-j} \mid U_j = u\}$, put

$$\xi_j(u) = \max_{v \in C_j(u)} \gamma\{\iota_j(u, v)\}, \quad \Psi_j = \frac{1}{|I_\varepsilon|} \int_{I_\varepsilon} \xi_j(u) du, \quad \Delta_j = \gamma_U^* - \Psi_j, \quad (2)$$

and

$$\mathcal{D}_U = \{j \leq p : \Delta_j > 0\}. \quad (3)$$

The projection theorem applied to the trivial sigma-field shows that the unconditional law of Y has tail index γ_U^* . Thus the baseline in (??) can be estimated directly from the response sample; no zero-gap coordinate is needed.

Let

$$Q_0(s) = F_Y^{-1}(1 - s^{-1}) = s^{\gamma_U^*} \ell_0(s), \quad s > 1, \quad (4)$$

and, for $0 < a \leq 1/3$, define the unconditional tail remainder

$$b_0(a) = \sup_{s \geq (3a)^{-1}} \sup_{t > 1} \frac{|\log\{\ell_0(ts)/\ell_0(s)\}|}{\log t}. \quad (5)$$

The upper bound $\gamma_U^* \leq \Gamma$ is retained from the manuscript. Atoms are allowed throughout; the randomized conditional distributional transform supplies the uniform probability-integral-transform variables used below.

1.2 The two finite-sample ingredients

The first ingredient is the score-level concentration theorem already proved for the original sliding-window statistic. We record it in a form convenient for post-selection inference. Let Λ be a finite collection of score tunings $\lambda = (\alpha_\lambda, h_\lambda)$. For each λ , let $\hat{\Psi}_j^\lambda$ be the original empirical-rank sliding score and let $L_n = |\{r : r/(n+1) \in I_\varepsilon\}|$. Put

$$\begin{aligned} d_\lambda &= \lfloor h_\lambda(n+1) \rfloor, & \chi_\lambda &= 2d_\lambda + 1, \\ W_{\lambda r} &= \{q \in \{1, \dots, n\} : |q - r| \leq d_\lambda\}, & m_{\lambda r} &= |W_{\lambda r}|, \\ k_{\lambda r} &= \lfloor \alpha_\lambda m_{\lambda r} \rfloor, & K_\lambda &= \min_{r: r/(n+1) \in I_\varepsilon} k_{\lambda r}, \end{aligned} \quad (6)$$

$$v_\lambda = \frac{\chi_\lambda}{L_n^2} \sum_{r: r/(n+1) \in I_\varepsilon} \frac{1}{k_{\lambda r}}, \quad c_\lambda = \frac{\chi_\lambda}{L_n K_\lambda}, \quad \rho_\lambda(x) = \sqrt{2v_\lambda x} + c_\lambda x. \quad (7)$$

Choose $\eta_\lambda > 0$, $x_\lambda > 0$, and $0 < \tau_\lambda \leq 3\alpha_\lambda$. Suppose deterministic certificates $\bar{q}_\lambda, \bar{\omega}_\lambda, \bar{b}_\lambda$ satisfy

$$\max_{j \leq p} \left| \frac{1}{L_n} \sum_{r: r/(n+1) \in I_\varepsilon} \xi_j\{r/(n+1)\} - \Psi_j \right| \leq \bar{q}_\lambda, \quad (8)$$

$$\omega_{n,\lambda}^\uparrow \{h_\lambda + \eta_\lambda + (n+1)^{-1}, \tau_\lambda, \alpha_\lambda\} \leq \bar{\omega}_\lambda, \quad (9)$$

$$b_{n,\lambda}(3\alpha_\lambda) \leq \bar{b}_\lambda. \quad (10)$$

Then the completed sliding-score argument gives

$$\mathbb{P}\left(\max_{j \leq p} |\hat{\Psi}_j^\lambda - \Psi_j| > r_\lambda\right) \leq \pi_\lambda, \quad (11)$$

where

$$r_\lambda = \bar{q}_\lambda + 2\bar{\omega}_\lambda + \bar{b}_\lambda + (\Gamma + \bar{b}_\lambda)\rho_\lambda(x_\lambda), \quad (12)$$

$$\pi_\lambda = 2pe^{-2n\eta_\lambda^2} + pn\tau_\lambda + pL_ne^{-(K_\lambda+1)/4} + 4pe^{-x_\lambda}. \quad (13)$$

No independence across coordinates or across tunings is asserted or needed. The role of (??)–(??) is solely to turn the already proved theoretical radius into an honest numerical certificate.

The second ingredient, which is new, is a finite-sample band for the global baseline.

Lemma 1.1 (Finite-sample global-Hill band). *Let $Y_1, \dots, Y_n$ be iid with upper quantile (??). Fix $0 < a \leq 1/3$, put $k = \lfloor an \rfloor \geq 1$, and define*

$$\hat{\gamma}_0(a) = \frac{1}{k} \sum_{r=1}^k \log \frac{Y_{(n-r+1)}}{Y_{(n-k)}}. \quad (14)$$

For every $0 < t \leq 1$,

$$\mathbb{P}\{|\hat{\gamma}_0(a) - \gamma_U^*| > s_0(a, t)\} \leq 2e^{-kt^2/4} + e^{-(k+1)/4}, \quad (15)$$

where

$$s_0(a, t) = \Gamma t + (1+t)b_0(a). \quad (16)$$

Equivalently, if $x \leq k/4$ and $t = 2\sqrt{x/k}$, then the first probability term on the right of (??) is $2e^{-x}$.

Proof. Use the randomized distributional transform to construct iid standard uniform variables $V_1, \dots, V_n$ such that $Y_i = Q_0(V_i^{-1})$ almost surely. Let $V_{(1)} < \dots < V_{(n)}$ be their order statistics and set

$$G_k = \frac{1}{k} \sum_{r=1}^k \log \frac{V_{(k+1)}}{V_{(r)}}. \quad (17)$$

The Rényi representation gives that G_k is the mean of k iid standard exponentials. Therefore

$$\mathbb{P}(|G_k - 1| > t) \leq 2e^{-kt^2/4}, \quad 0 < t \leq 1. \quad (18)$$

Moreover, the binomial Chernoff bound for the $(k+1)$ st uniform order statistic gives

$$\mathbb{P}\{V_{(k+1)} > 3a\} \leq e^{-(k+1)/4}. \quad (19)$$

On the event in which $V_{(k+1)} \leq 3a$, quantile monotonicity and (??) yield the exact decomposition

$$\hat{\gamma}_0(a) = \gamma_U^* G_k + R_k, \quad (20)$$

where

$$R_k = \frac{1}{k} \sum_{r=1}^k \log \frac{\ell_0(V_{(r)}^{-1})}{\ell_0(V_{(k+1)}^{-1})}. \quad (21)$$

For every summand, the base point satisfies $V_{(k+1)}^{-1} \geq (3a)^{-1}$ and the multiplicative increment is $V_{(k+1)}/V_{(r)} > 1$. Hence (??) implies

$$|R_k| \leq b_0(a)G_k. \quad (22)$$

On $|G_k - 1| \leq t$,

$$\begin{aligned} |\hat{\gamma}_0(a) - \gamma_U^*| &\leq \gamma_U^*|G_k - 1| + |R_k| \\ &\leq \Gamma t + (1+t)b_0(a). \end{aligned}$$

The union of the two exceptional events in (??) and (??) proves (??). $\square$

1.3 Simultaneous confidence intervals and coordinatewise oracle tuning

Let $\mathcal{K}$ be a finite collection of global tail fractions. For $\kappa \in \mathcal{K}$, write a_κ for the fraction, $k_\kappa = \lfloor a_\kappa n \rfloor$, choose $0 < t_\kappa \leq 1$, and put

$$\hat{\gamma}^\kappa = \hat{\gamma}_0(a_\kappa), \quad s_\kappa = \Gamma t_\kappa + (1+t_\kappa)b_0(a_\kappa), \quad \pi_\kappa^0 = 2e^{-k_\kappa t_\kappa^2/4} + e^{-(k_\kappa+1)/4}. \quad (23)$$

The score and baseline estimators may all be computed from the same sample.

Theorem 1.2 (Post-selection simultaneous gap inference). *Assume (??) for every $\lambda \in \Lambda$ and Lemma ?? for every $\kappa \in \mathcal{K}$. Define*

$$\Pi = \sum_{\lambda \in \Lambda} \pi_\lambda + \sum_{\kappa \in \mathcal{K}} \pi_\kappa^0. \quad (24)$$

For $j \leq p$, set

$$\underline{\Delta}_j = \left[\max_{\lambda \in \Lambda, \kappa \in \mathcal{K}} \left\{ \hat{\gamma}^\kappa - \hat{\Psi}_j^\lambda - s_\kappa - r_\lambda \right\} \right]_+, \quad (25)$$

$$\overline{\Delta}_j = \left[\min_{\lambda \in \Lambda, \kappa \in \mathcal{K}} \left\{ \hat{\gamma}^\kappa - \hat{\Psi}_j^\lambda + s_\kappa + r_\lambda \right\} \right]_+. \quad (26)$$

Then

$$\mathbb{P} \{ \underline{\Delta}_j \leq \Delta_j \leq \overline{\Delta}_j \text{ for every } j \leq p \} \geq 1 - \Pi. \quad (27)$$

Consequently, for every deterministic threshold $z \geq 0$, with probability at least $1 - \Pi$,

$$\{j : \underline{\Delta}_j > z\} \subseteq \{j : \Delta_j > z\} \subseteq \{j : \overline{\Delta}_j > z\}. \quad (28)$$

The maximizer in (??) may be selected after seeing the entire data set and may be different for different coordinates.

Proof. Let

$$\mathcal{E} = \bigcap_{\lambda \in \Lambda} \left\{ \max_{j \leq p} |\hat{\Psi}_j^\lambda - \Psi_j| \leq r_\lambda \right\} \cap \bigcap_{\kappa \in \mathcal{K}} \{ |\hat{\gamma}^\kappa - \gamma_U^*| \leq s_\kappa \}. \quad (29)$$

A union bound gives $\mathbb{P}(\mathcal{E}) \geq 1 - \Pi$; no independence is used. On $\mathcal{E}$, for every j, λ, κ ,

$$\hat{\gamma}^\kappa - \hat{\Psi}_j^\lambda - (s_\kappa + r_\lambda) \leq \gamma_U^* - \Psi_j = \Delta_j, \quad (30)$$

$$\hat{\gamma}^\kappa - \hat{\Psi}_j^\lambda + (s_\kappa + r_\lambda) \geq \gamma_U^* - \Psi_j = \Delta_j. \quad (31)$$

Taking the maximum of the lower endpoints and the minimum of the upper endpoints preserves the two inequalities. Since $\Delta_j \geq 0$, applying the positive-part map does not alter coverage. This proves (??). The set inclusions in (??) are deterministic consequences of simultaneous coverage. $\square$

Corollary 1.3 (Finite-sample FWER, sure coverage, exact recovery, and an oracle property). Define the inner and outer data-driven sets

$$\widehat{\mathcal{D}}_{\text{in}} = \{j : \underline{\Delta}_j > 0\}, \quad \widehat{\mathcal{D}}_{\text{out}} = \{j : \overline{\Delta}_j > 0\}, \quad (32)$$

and their data-driven cardinalities $\widehat{d}_{\text{in}} = |\widehat{\mathcal{D}}_{\text{in}}|$ and $\widehat{d}_{\text{out}} = |\widehat{\mathcal{D}}_{\text{out}}|$. Then

$$\mathbb{P}\{\widehat{\mathcal{D}}_{\text{in}} \not\subseteq \mathcal{D}_U\} \leq \Pi, \quad (33)$$

$$\mathbb{P}\{\mathcal{D}_U \not\subseteq \widehat{\mathcal{D}}_{\text{out}}\} \leq \Pi. \quad (34)$$

Thus the inner set controls the familywise probability of even one false positive, while the outer set is an honest sure-screening set. Neither rule uses $|\mathcal{D}_U|$, $\Delta_{\min}$, or a common good tuning.

Moreover, on the event (??), a coordinate $j \in \mathcal{D}_U$ is confirmed by the inner set whenever there exists a pair $(\lambda_j, \kappa_j) \in \Lambda \times \mathcal{K}$ such that

$$\Delta_j > 2\{r_{\lambda_j} + s_{\kappa_j}\}. \quad (35)$$

If (??) holds for every $j \in \mathcal{D}_U$, then

$$\mathbb{P}(\widehat{\mathcal{D}}_{\text{in}} = \mathcal{D}_U) \geq 1 - \Pi. \quad (36)$$

Different positive-gap coordinates may use different oracle pairs. In particular, $\widehat{d}_{\text{in}}$ consistently estimates $|\mathcal{D}_U|$ under this separation condition.

Proof. For $j \notin \mathcal{D}_U$, $\Delta_j = 0$. Simultaneous coverage implies $\underline{\Delta}_j \leq 0$, while by definition $\underline{\Delta}_j \geq 0$, so $\underline{\Delta}_j = 0$. Hence no null coordinate is selected by the inner set on (??), proving (??). For $j \in \mathcal{D}_U$, simultaneous coverage gives $\overline{\Delta}_j \geq \Delta_j > 0$, hence $j \in \widehat{\mathcal{D}}_{\text{out}}$; this proves (??).

Fix $j \in \mathcal{D}_U$ and a pair satisfying (??). On (??),

$$\widehat{\gamma}^{\kappa_j} - \widehat{\Psi}_j^{\lambda_j} - s_{\kappa_j} - r_{\lambda_j} \geq \Delta_j - 2\{s_{\kappa_j} + r_{\lambda_j}\} > 0.$$

Therefore the maximum in (??) is positive and j is confirmed. If the condition holds for all positive-gap coordinates, there are neither false positives nor false negatives in the inner set on the good event, which proves (??). $\square$

Corollary 1.4 (Explicit level- δ calibration). Let $M = |\Lambda|$, $M_0 = |\mathcal{K}|$, and fix $0 < \delta < 1$. For every $\lambda \in \Lambda$, choose

$$\eta_\lambda = \sqrt{\frac{\log(16pM/\delta)}{2n}}, \quad \tau_\lambda = \frac{\delta}{8Mpn}, \quad x_\lambda = \log(32pM/\delta), \quad (37)$$

provided $\tau_\lambda \leq 3\alpha_\lambda$, and require

$$K_\lambda + 1 \geq 4\log(8MpL_n/\delta). \quad (38)$$

For every $\kappa \in \mathcal{K}$, choose

$$t_\kappa = 2\sqrt{\frac{\log(8M_0/\delta)}{k_\kappa}}, \quad (39)$$

provided $t_\kappa \leq 1$, and require

$$k_\kappa + 1 \geq 4\log(4M_0/\delta). \quad (40)$$

Then $\Pi \leq \delta$, so all conclusions of Theorem ?? and Corollary ?? hold with confidence at least $1 - \delta$.

Proof. For a fixed score tuning, the four terms in (??) are, respectively, at most $\delta/(8M)$, $\delta/(8M)$, $\delta/(8M)$, and $\delta/(8M)$ under (??)–(??). Thus $\sum_{\lambda} \pi_{\lambda} \leq \delta/2$. For a fixed global tuning, the two terms in (??) are each at most $\delta/(4M_0)$ under (??)–(??), so $\sum_{\kappa} \pi_{\kappa}^0 \leq \delta/2$. $\square$

Remark 1.5 (Why this is a genuine tuning theorem). The rule chooses (α, h) and the global tail fraction after seeing the data, coordinate by coordinate, by maximizing a lower confidence bound. The multiplicity cost is explicit in Π , and no stochastic independence across coordinates, tunings, or the baseline estimate is invoked. The structural bias certificates in (??)–(??) cannot be deleted uniformly over bare conditional regular variation: the delayed-tail-onset construction already proves that two models with a gap of order $1/\log n$ can have asymptotically indistinguishable samples. What is adaptive here is the tuning and the retained size, not an impossible adaptation to an unquantified tail onset.

Theorem 1.6 (Arbitrary pilot tuning by honest sample splitting). *Split the observations into an independent pilot sample $\mathcal{T}$ and an inference sample $\mathcal{I}$. The pilot may choose, by an arbitrary measurable rule, one global tuning $\hat{\kappa}$ and coordinate-specific score tunings $\hat{\lambda}_1, \dots, \hat{\lambda}_p$; it may also choose any preliminary retained size or ranking.*

Suppose that, conditionally on $\mathcal{T}$, the inference-sample estimators satisfy

$$\mathbb{P}\left(|\hat{\gamma}^{\hat{\kappa}} - \gamma_U^*| > s_{\hat{\kappa}} \mid \mathcal{T}\right) \leq \pi_0, \quad (41)$$

$$\mathbb{P}\left(|\hat{\Psi}_j^{\hat{\lambda}_j} - \Psi_j| > r_{j, \hat{\lambda}_j} \mid \mathcal{T}\right) \leq \pi_j, \quad j \leq p, \quad (42)$$

where the conditional failure budgets obey $\pi_0 + \sum_{j=1}^p \pi_j \leq \delta$ almost surely. Define

$$\underline{\Delta}_j^{\text{split}} = \left[\hat{\gamma}^{\hat{\kappa}} - \hat{\Psi}_j^{\hat{\lambda}_j} - s_{\hat{\kappa}} - r_{j, \hat{\lambda}_j} \right]_+. \quad (43)$$

Then

$$\mathbb{P}\left\{\{j : \underline{\Delta}_j^{\text{split}} > 0\} \not\subseteq \mathcal{D}_U\right\} \leq \delta. \quad (44)$$

If every $j \in \mathcal{D}_U$ satisfies, almost surely with respect to the pilot,

$$\Delta_j > 2\{s_{\hat{\kappa}} + r_{j, \hat{\lambda}_j}\}, \quad (45)$$

then the split selector equals $\mathcal{D}_U$ with probability at least $1 - \delta$. No cardinality penalty for the pilot search space appears.

Proof. Condition on $\mathcal{T}$. The selected tunings are then fixed. By the union bound, the event on which all inequalities in (??)–(??) hold has conditional probability at least $1 - \delta$. On that event, the deterministic argument of Corollary ?? applies coordinate by coordinate. Integrate the conditional probability with respect to the pilot sample. The power claim follows from the same inequality used in (??). $\square$

1.4 Temporal dependence by dependence-safe thinning

The preceding results permit arbitrary dependence among coordinates within one observation, but so far the rows are iid. A completely rigorous extension to absolutely regular time series is obtained by thinning. It sacrifices sample size, but it introduces no new local-Hill argument and makes the degradation explicit.

Let $Z_i = (U_i, Y_i)$ be a strictly stationary sequence with one-observation law P . Write $\mathcal{F}_t = \sigma(Z_s : s \leq t)$ and define the absolute-regularity coefficient, with the total-variation convention used below, by

$$\beta(q) = \sup_t \mathbb{E} [\text{TV}(\mathcal{L}((Z_{t+q}, Z_{t+q+1}, \dots) \mid \mathcal{F}_t), \mathcal{L}(Z_{t+q}, Z_{t+q+1}, \dots))] . \quad (46)$$

Lemma 1.7 (Product approximation for a thinned absolutely regular sample). *For an integer $q \geq 1$, put*

$$t_r = 1 + (r-1)q, \quad N = 1 + \left\lfloor \frac{n-1}{q} \right\rfloor, \quad 1 \leq r \leq N. \quad (47)$$

Then

$$\text{TV}(\mathcal{L}(Z_{t_1}, \dots, Z_{t_N}), P^{\otimes N}) \leq (N-1)\beta(q). \quad (48)$$

Consequently, for every event A measurable with respect to the thinned sample,

$$|\mathbb{P}\{(Z_{t_1}, \dots, Z_{t_N}) \in A\} - P^{\otimes N}(A)| \leq (N-1)\beta(q). \quad (49)$$

Proof. We first record a chain inequality. Let $X_1, \dots, X_N$ have common marginal P , and let $K_r(x_1, \dots, x_{r-1}; \cdot)$ be a regular conditional law of X_r given its predecessors. If $\mu_r = \mathcal{L}(X_1, \dots, X_r)$, then

$$\text{TV}(\mu_r, P^{\otimes r}) \leq \text{TV}\{\mu_{r-1}K_r, \mu_{r-1}P\} + \text{TV}\{\mu_{r-1}P, P^{\otimes(r-1)}P\} \quad (50)$$

$$\leq \int \text{TV}\{K_r(x_{1:r-1}; \cdot), P\} \mu_{r-1}(dx_{1:r-1}) + \text{TV}(\mu_{r-1}, P^{\otimes(r-1)}). \quad (51)$$

The second inequality follows directly from the variational definition of TV: adjoining a common product kernel preserves total variation, while integrating two kernels against the same measure bounds total variation by the average of their pointwise distances. Iterating (??) yields

$$\text{TV}\{\mathcal{L}(X_1, \dots, X_N), P^{\otimes N}\} \leq \sum_{r=2}^N \mathbb{E} [\text{TV}\{\mathcal{L}(X_r \mid X_1, \dots, X_{r-1}), P\}]. \quad (52)$$

Apply this with $X_r = Z_{t_r}$. The sigma-field generated by $X_1, \dots, X_{r-1}$ is contained in $\mathcal{F}_{t_{r-1}}$, and $t_r - t_{r-1} = q$. Conditional expectation of probability kernels and convexity of total variation give

$$\begin{aligned} & \mathbb{E} [\text{TV}\{\mathcal{L}(Z_{t_r} \mid Z_{t_1}, \dots, Z_{t_{r-1}}), P\}] \\ & \leq \mathbb{E} [\text{TV}\{\mathcal{L}(Z_{t_r} \mid \mathcal{F}_{t_{r-1}}), P\}] \leq \beta(q). \end{aligned}$$

The last inequality follows from (??) by projecting the future sequence onto its first coordinate, because measurable maps cannot increase total variation. Substitution in (??) proves (??); (??) is the defining event bound for total variation. $\square$

Theorem 1.8 (Simultaneous inference under temporal dependence). *Assume the one-observation law P satisfies the population and projected-tail conditions used in Theorem ???. Compute every global Hill estimate, every sliding rank score, and the confidence bounds (??)–(??) using only the thinned observations in (??), with N replacing n in all radii and rank geometry. Let Π_N be the resulting iid failure bound. Then*

$$\mathbb{P}\{\underline{\Delta}_j \leq \Delta_j \leq \overline{\Delta}_j \text{ for every } j \leq p\} \geq 1 - \Pi_N - (N-1)\beta(q). \quad (53)$$

Every FWER, sure-coverage, oracle, and exact-recovery conclusion of Corollary ?? remains valid after replacing Π by $\Pi_N + (N-1)\beta(q)$.

Proof. Under the product law $P^{\otimes N}$, the entire thinned-sample construction is exactly the iid construction of Theorem ??, so its good event has probability at least $1 - \Pi_N$. The probability of that same measurable event under the dependent thinned sample differs by at most $(N - 1)\beta(q)$ by Lemma ?. Once the good event holds, all selection conclusions are deterministic and therefore transfer with the same additive error. $\square$

Corollary 1.9 (Explicit effective sample sizes). *If $\beta(q) \leq Ce^{-cq}$, taking $q_n = \lceil 2c^{-1} \log n \rceil$ gives*

$$N_n \asymp \frac{n}{\log n}, \quad (N_n - 1)\beta(q_n) = O\{(n \log n)^{-1}\}. \quad (54)$$

If $\beta(q) \leq Cq^{-r}$ for some $r > 0$, taking

$$q_n = \left\lceil n^{1/(r+1)} \ell_n \right\rceil, \quad \ell_n \longrightarrow \infty, \quad \ell_n = o\{n^{r/(r+1)}\}, \quad (55)$$

gives

$$N_n \asymp \frac{n^{r/(r+1)}}{\ell_n}, \quad (N_n - 1)\beta(q_n) = O(\ell_n^{-(r+1)}). \quad (56)$$

Thus every iid occurrence of $n\alpha$ or $n\alpha h$ in the stochastic and local-feasibility conditions is replaced by $N_n\alpha$ or $N_n\alpha h$.

Proof. For geometric mixing, $e^{-cq_n} \leq n^{-2}$ and $N_n \leq n/q_n + 1$, which gives (?). For polynomial mixing,

$$(N_n - 1)\beta(q_n) \leq C \frac{n}{q_n^{r+1}} = O(\ell_n^{-(r+1)}),$$

and $N_n \asymp n/q_n$ gives the displayed effective sample size. $\square$

1.5 Contamination: breakdown and the sharp ϵ/α floor

We now distinguish the trimming constant ε in I_ε from a rowwise contamination fraction, denoted ϵ_c .

Proposition 1.10 (One-row replacement breakdown of the sliding Hill score). *Fix a coordinate j and a realized sample for which at least one trimmed rank window is valid. Keep the covariates fixed and replace a single response Y_i by M . If observation i belongs to at least one valid trimmed window, then*

$$\widehat{\Psi}_j(M) \longrightarrow +\infty \quad (M \rightarrow \infty). \quad (57)$$

Consequently the finite-sample replacement breakdown point of every nondegenerate sliding Hill score is at most $1/n$.

Proof. Let $\mathcal{R}_i$ be the nonempty set of trimmed grid centres whose rank windows contain observation i . For a centre $r \in \mathcal{R}_i$, let $k_r \geq 1$ be its number of upper order statistics. Once M exceeds every unchanged response in that window, the replaced response is its largest order statistic, while the other upper order statistics and the threshold are independent of M . Hence

$$\widehat{\xi}_j\{r/(n+1); M\} = \frac{\log M}{k_r} + C_r, \quad (58)$$

where C_r is finite and does not depend on M . Averaging over all trimmed centres gives

$$\widehat{\Psi}_j(M) = \left\{ \frac{1}{L_n} \sum_{r \in \mathcal{R}_i} \frac{1}{k_r} \right\} \log M + C, \quad (59)$$

with a strictly positive coefficient. This proves (?). $\square$

The next lemma gives the exact geometry of Huber contamination neighborhoods. For a probability measure P and $0 \leq \epsilon < 1$, write

$$\mathcal{H}_\epsilon(P) = \{(1 - \epsilon)P + \epsilon R : R \text{ is a probability measure}\}. \quad (60)$$

Lemma 1.11 (Intersection of Huber neighborhoods). *For probability measures P, Q ,*

$$\mathcal{H}_\epsilon(P) \cap \mathcal{H}_\epsilon(Q) \neq \emptyset \iff \text{TV}(P, Q) \leq \frac{\epsilon}{1 - \epsilon}. \quad (61)$$

Here $\text{TV}(P, Q) = \sup_A |P(A) - Q(A)|$.

Proof. Let $\mu = P + Q$ and write $p = dP/d\mu$, $q = dQ/d\mu$. The least finite measure that dominates both $(1 - \epsilon)P$ and $(1 - \epsilon)Q$ has density $(1 - \epsilon) \max(p, q)$ and total mass

$$(1 - \epsilon) \int \max(p, q) d\mu = (1 - \epsilon) \{1 + \text{TV}(P, Q)\}. \quad (62)$$

If a probability measure M belongs to both Huber neighborhoods, then M dominates both subprobability measures, so the quantity in (??) is at most one. This proves necessity.

Conversely, suppose that the quantity in (??) is at most one. Let M_0 be the measure with density $(1 - \epsilon) \max(p, q)$ and add an arbitrary nonnegative measure of mass $1 - M_0(\Omega)$ to obtain a probability measure M . Then $M - (1 - \epsilon)P$ and $M - (1 - \epsilon)Q$ are nonnegative and each has total mass ϵ . Dividing them by ϵ gives probability measures R_P, R_Q such that

$$M = (1 - \epsilon)P + \epsilon R_P = (1 - \epsilon)Q + \epsilon R_Q.$$

Thus the two neighborhoods intersect. $\square$

We use the exact tail submodel already embedded in the screening class. Fix $q \in (0, 1/2)$, $\gamma_0 > 0$, and $0 < \delta \leq \gamma_0/2$, and put $T = q^{-\gamma_0}$. Let $U \sim \text{Unif}([0, 1]^p)$, $B \sim \text{Bernoulli}(q)$, and $E \sim \text{Exp}(1)$ be independent. If $B = 0$, draw Y from one fixed continuous law supported in $(T/4, T/2)$. If $B = 1$, define

$$P_0 : Y = T \exp(\gamma_0 E), \quad (63)$$

$$P_j : Y = T \exp\{(\gamma_0 - \delta U_j)E\}, \quad 1 \leq j \leq p. \quad (64)$$

For fixed parameters, these laws satisfy the primitive tail assumptions and full fibre support. Along triangular arrays, the usual uniform coefficient bound holds whenever $\delta \log(1/q)/\gamma_0 = O(1)$. Under P_j ,

$$\gamma(u) = \gamma_0 - \delta u_j, \quad \gamma_U^* = \gamma_0, \quad \Delta_j = \delta/2, \quad \Delta_\ell = 0 \quad (\ell \neq j). \quad (65)$$

Theorem 1.12 (Infinite-sample contamination lower bound). *Let $0 \leq \epsilon_c < 1/2$. If*

$$\frac{q\delta}{2\gamma_0} \leq \frac{\epsilon_c}{1 - \epsilon_c}, \quad (66)$$

then

$$\mathcal{H}_{\epsilon_c}(P_0) \cap \mathcal{H}_{\epsilon_c}(P_j) \neq \emptyset. \quad (67)$$

Consequently, for every sample size n and every possibly randomized selector $\hat{S}$ based on the full contaminated sample, there exist $\tilde{P}_0 \in \mathcal{H}_{\epsilon_c}(P_0)$ and $\tilde{P}_j \in \mathcal{H}_{\epsilon_c}(P_j)$ such that

$$\tilde{P}_0^{\otimes n} \{j \in \hat{S}\} + \tilde{P}_j^{\otimes n} \{j \notin \hat{S}\} = 1. \quad (68)$$

In terms of the positive gap, the nonidentifiable region contains

$$\Delta_j \leq \frac{\gamma_0 \epsilon_c}{q(1 - \epsilon_c)}. \quad (69)$$

Thus, when q is proportional to the working tail fraction α , the contamination floor is of order ϵ_c/α and persists even with infinitely many observations.

Proof. The laws P_0 and P_j coincide on $\{B = 0\}$ and have the same laws of (U, B) . Conditional on $B = 1$ and $U = u$, the excess $X = \log(Y/T)$ is exponential with mean γ_0 under P_0 and mean $\theta(u) = \gamma_0 - \delta u_j$ under P_j . Therefore

$$\text{TV}(P_j, P_0) = q \mathbb{E} [\text{TV}\{\text{Exp}(\theta(U)), \text{Exp}(\gamma_0)\}]. \quad (70)$$

For $z = \delta u_j / \gamma_0 \in [0, 1/2]$, the Kullback–Leibler divergence is

$$\begin{aligned} D\{\text{Exp}(\gamma_0(1 - z)) \parallel \text{Exp}(\gamma_0)\} &= -\log(1 - z) - z \\ &= \sum_{m=2}^{\infty} \frac{z^m}{m} \leq \frac{z^2}{2(1 - z)} \leq z^2. \end{aligned} \quad (71)$$

Pinsker's inequality gives a total variation distance at most $z/\sqrt{2}$, and hence at most z . Using $\mathbb{E}U_j = 1/2$ in (??),

$$\text{TV}(P_j, P_0) \leq \frac{q\delta}{2\gamma_0}. \quad (72)$$

Condition (??) and Lemma ?? prove (??). Choose a common law $M \in \mathcal{H}_{\epsilon_c}(P_0) \cap \mathcal{H}_{\epsilon_c}(P_j)$ and set $\tilde{P}_0 = \tilde{P}_j = M$. The distribution of the selector is then identical under the two experiments. If its common probability of including j is r , the two terms in (??) are r and $1 - r$. Finally, substitute $\delta = 2\Delta_j$ into (??) to obtain (??). $\square$

Corollary 1.13 (Strong-replacement lower bound). *Fix $0 < \epsilon_c < 1$ and let a strong adversary be allowed to replace at most $m_n = \lceil n\epsilon_c \rceil$ complete rows. If*

$$\frac{q\delta}{2\gamma_0} \leq \frac{\epsilon_c}{2 - \epsilon_c}, \quad (73)$$

then for every selector $\hat{S}$ there are possibly randomized legal replacement strategies under P_0 and P_j whose induced laws $Q_{0,n}^{\text{adv}}$ and $Q_{j,n}^{\text{adv}}$ satisfy

$$Q_{0,n}^{\text{adv}}\{j \in \hat{S}\} + Q_{j,n}^{\text{adv}}\{j \notin \hat{S}\} \geq 1 - 2\exp(-n\epsilon_c/6). \quad (74)$$

Equivalently, strong replacement contamination makes gaps

$$\Delta_j \leq \frac{\gamma_0 \epsilon_c}{q(2 - \epsilon_c)} \quad (75)$$

nonidentifiable up to an error tending exponentially to zero.

Proof. Set $\epsilon_0 = \epsilon_c/2$. By (??) and (??),

$$\text{TV}(P_j, P_0) \leq \frac{\epsilon_0}{1 - \epsilon_0}.$$

Lemma ?? therefore supplies a common probability measure M and probability measures R_0, R_j such that

$$M = (1 - \epsilon_0)P_0 + \epsilon_0 R_0 = (1 - \epsilon_0)P_j + \epsilon_0 R_j. \quad (76)$$

Under clean law P_r , $r \in \{0, j\}$, generate iid clean rows $X_i \sim P_r$, iid indicators $C_i \sim \text{Bernoulli}(\epsilon_0)$, and iid replacement rows $R_i \sim R_r$, all independent. Let $N = \sum_i C_i$. If $N \leq m_n$, replace precisely the rows with $C_i = 1$ by R_i ; if $N > m_n$, make no replacements. This strategy is legal on every sample path. Without the truncation on N , the output rows would be iid with law M by (?). Coupling the truncated and untruncated outputs on the same variables therefore gives

$$\text{TV}(Q_{r,n}^{\text{adv}}, M^{\otimes n}) \leq \mathbb{P}(N > m_n). \quad (77)$$

Here $\mathbb{E}N = n\epsilon_0 = n\epsilon_c/2$ and $m_n \geq 2\mathbb{E}N$. The multiplicative Chernoff inequality yields

$$\mathbb{P}(N > m_n) \leq \mathbb{P}(N > 2\mathbb{E}N) \leq \exp(-\mathbb{E}N/3) = \exp(-n\epsilon_c/6). \quad (78)$$

The triangle inequality gives $\text{TV}(Q_{0,n}^{\text{adv}}, Q_{j,n}^{\text{adv}}) \leq 2e^{-n\epsilon_c/6}$. The testing inequality $Q_0(A) + Q_1(A^c) \geq 1 - \text{TV}(Q_0, Q_1)$, with $A = \{j \in \hat{S}\}$, proves (?). Finally substitute $\delta = 2\Delta_j$ into (?). $\square$

The lower bound is attained in order on the same submodel by a bounded statistic. The next theorem uses the stronger deterministic replacement model: an adversary may inspect a clean iid sample and replace at most $\lfloor n\epsilon_c \rfloor$ complete rows by arbitrary points of $[0, 1]^p \times (0, \infty)$.

Theorem 1.14 (Replacement-robust matching upper bound). *Let a clean sample be iid from P_j in (?). After at most $\lfloor n\epsilon_c \rfloor$ arbitrary row replacements, denote the observations by $(\tilde{U}_i, \tilde{Y}_i)$. Define, for $\ell \leq p$,*

$$S_\ell = \frac{1}{n} \sum_{i=1}^n \left(\tilde{U}_{i\ell} - \frac{1}{2} \right) \mathbf{1}_{\{\tilde{Y}_i > Te^{\gamma_0}\}}, \quad \hat{J} \in \arg \min_{\ell \leq p} S_\ell. \quad (79)$$

For $x > 0$, put

$$r_n(x) = \sqrt{\frac{qx}{2n}} + \frac{x}{3n}. \quad (80)$$

If

$$\frac{qe^{-2}}{12} \frac{\delta}{\gamma_0} > 2\{\epsilon_c + r_n(x)\}, \quad (81)$$

then, uniformly over all such replacement strategies,

$$\mathbb{P}_j(\hat{J} \neq j) \leq 2pe^{-x}. \quad (82)$$

In particular, with $x = \log(2p/\eta)$, exact recovery has probability at least $1 - \eta$ whenever

$$\frac{\delta}{\gamma_0} > 24e^2 \left\{ \frac{\epsilon_c}{q} + \sqrt{\frac{\log(2p/\eta)}{2nq}} + \frac{\log(2p/\eta)}{3nq} \right\}. \quad (83)$$

Proof. First work with the clean sample and define

$$Z_{i\ell} = \left(U_{i\ell} - \frac{1}{2} \right) \mathbf{1}_{\{Y_i > Te^{\gamma_0}\}}, \quad \mu_\ell = \mathbb{E}_j Z_{i\ell}. \quad (84)$$

For $\ell \neq j$, the event in the indicator depends only on (B, E, U_j) and is independent of U_ℓ , so

$$\mu_\ell = 0, \quad \ell \neq j. \quad (85)$$

For the active coordinate, write $a = \delta/\gamma_0 \in (0, 1/2]$ and

$$f_a(u) = \exp \left\{ -\frac{1}{1-au} \right\}. \quad (86)$$

Conditional on $B = 1$ and $U_j = u$, the event $Y > Te^{\gamma_0}$ is equivalent to $E > \gamma_0/(\gamma_0 - \delta u)$, so

$$\mu_j = q \int_0^1 \left(u - \frac{1}{2} \right) f_a(u) du. \quad (87)$$

Now

$$f'_a(u) = -\frac{a}{(1-au)^2} \exp \left\{ -\frac{1}{1-au} \right\} \leq -ae^{-2}, \quad 0 \leq u \leq 1, \quad (88)$$

because $1/(1-au) \leq 2$. If U, U' are iid uniform variables, then

$$\begin{aligned} \int_0^1 \left(u - \frac{1}{2} \right) f_a(u) du &= \text{Cov}\{U, f_a(U)\} \\ &= \frac{1}{2} \mathbb{E}[(U - U')\{f_a(U) - f_a(U')\}] \\ &\leq -\frac{ae^{-2}}{2} \mathbb{E}(U - U')^2 = -\frac{ae^{-2}}{12}. \end{aligned} \quad (89)$$

Thus

$$\mu_j \leq -\frac{qe^{-2}}{12} \frac{\delta}{\gamma_0}. \quad (90)$$

For every ℓ , $Z_{i\ell} \in [-1/2, 1/2]$ and

$$\text{Var}(Z_{i\ell}) \leq \mathbb{E}Z_{i\ell}^2 \leq \frac{1}{4} \mathbb{P}_j(Y > Te^{\gamma_0}) \leq \frac{q}{4}. \quad (91)$$

Also $|Z_{i\ell} - \mu_\ell| \leq 1$. Bernstein's inequality therefore gives

$$\mathbb{P}_j \left\{ \left| \frac{1}{n} \sum_{i=1}^n Z_{i\ell} - \mu_\ell \right| > r_n(x) \right\} \leq 2e^{-x}. \quad (92)$$

A union bound over $\ell \leq p$ shows that, outside probability $2pe^{-x}$,

$$\max_{\ell \leq p} \left| \frac{1}{n} \sum_{i=1}^n Z_{i\ell} - \mu_\ell \right| \leq r_n(x). \quad (93)$$

Each summand in (??) lies in $[-1/2, 1/2]$. Replacing one row can therefore change any coordinate average by at most $1/n$, and replacing at most $n\epsilon_c$ rows changes it by at most ϵ_c . Hence, on (??), every replacement strategy satisfies

$$S_j \leq -\frac{qe^{-2}}{12} \frac{\delta}{\gamma_0} + r_n(x) + \epsilon_c, \quad (94)$$

$$S_\ell \geq -r_n(x) - \epsilon_c, \quad \ell \neq j. \quad (95)$$

Condition (??) makes the first upper bound strictly smaller than every null lower bound, so $\hat{J} = j$ for every adversarial replacement on the clean event. This proves (??). Substituting $x = \log(2p/\eta)$ and rearranging (??) gives (??). $\square$

Corollary 1.15 (Sharp stochastic-contamination transition on the embedded submodel). *Take $q = 3\alpha$ and consider one-sparse exact recovery under strong row replacement contamination. Work in the admissible triangular-array regime*

$$0 < \Delta \leq \gamma_0/4, \quad \frac{\Delta}{\gamma_0} \log(1/\alpha) = O(1), \quad (96)$$

which keeps the tail coefficient of the embedded primitive model uniformly bounded. Along sequences with $n\epsilon_c \rightarrow \infty$, combining Corollary ??, the list-Fano lower bound for the same clean submodel, and Theorem ??, the minimax separation scale is, up to universal constants and the lower-order Bernstein term,

$$\frac{\Delta}{\gamma_0} \asymp \frac{\epsilon_c}{\alpha} + \sqrt{\frac{\log p}{n\alpha}}. \quad (97)$$

The first term is a contamination identification floor; the second is the clean high-dimensional stochastic boundary. Neither contains the bandwidth h .

Proof. Corollary ?? gives the necessary scale $\Delta/\gamma_0 \gtrsim \epsilon_c/q$; its testing lower bound tends to one under $n\epsilon_c \rightarrow \infty$. The list-Fano theorem gives the separate necessary scale $\Delta/\gamma_0 \gtrsim \sqrt{\log p/(nq)}$ for one-sparse exact recovery. Their maximum is equivalent, within a factor two, to their sum. Theorem ??, with $\delta = 2\Delta$, succeeds when that sum, plus $\log p/(nq)$, is smaller than a universal constant times Δ/γ_0 . Since $q = 3\alpha$, this proves (??) up to constants. $\square$

1.6 What these results close

The confidence construction supplies simultaneous, finite-sample inference for all p corrected gaps, a familywise-valid inner set, an honest sure-screening outer set, a data-driven retained size, and coordinate-specific tuning with an oracle guarantee. It does not need a zero-gap coordinate because the unconditional response tail estimates γ_U^* directly. Arbitrary cross-coordinate dependence and arbitrary dependence across the candidate tunings are harmless for validity. Under temporal absolute regularity, thinning transfers the whole construction with the explicit additive error $(N-1)\beta(q)$ and replaces the iid sample size by N .

The tuning statements are honest only for candidate estimators carrying valid fixed-tuning radii, including their tail, spatial, and quadrature bias certificates. The maximization of lower confidence bounds and the pilot choice are fully data driven; uniform adaptation to a completely unquantified tail onset is not claimed, because the delayed-onset impossibility theorem rules it out over bare (C1)–(C2).

The contamination results show both sides of the robustness question. The implemented sliding Hill score has one-row replacement breakdown. More fundamentally, no method can beat an ϵ_c/α gap floor under Huber or strong replacement contamination, while a bounded tail statistic attains that order, together with the clean $\sqrt{\log p/(n\alpha)}$ boundary, on the exact tail submodel. Extending the matching robust upper bound from that submodel to the full projected-quantile class requires a robust local threshold construction; that problem is not resolved here.